

Client: BrightStone Jewelers
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Gemstone Analytics Inc. Analysis of Diamond Pricing



Business Context

- BrightStone Jewelers sells thousands of unique diamonds online
- Large variation in carat, cut, color, clarity, and proportions
- Mispricing impacts revenue and conversion
- Goal: understand what drives price + improve pricing strategy**
- Key Insights
 - Carat overwhelmingly drives diamond prices, with clarity, color, and physical dimensions adding structured premiums, while depth and table contribute almost nothing.

Dataset Overview

- Dataset: Diamonds dataset (59,340 records) from internal BrightStone Jewelers data.
- Average Price: \$3,933
- Average Carat: 0.80 ct.

Variables

Carat(ct.)

Price(\$)

Clarity(IF-I1)

Cut (Ideal-Fair)

Color(D-J)

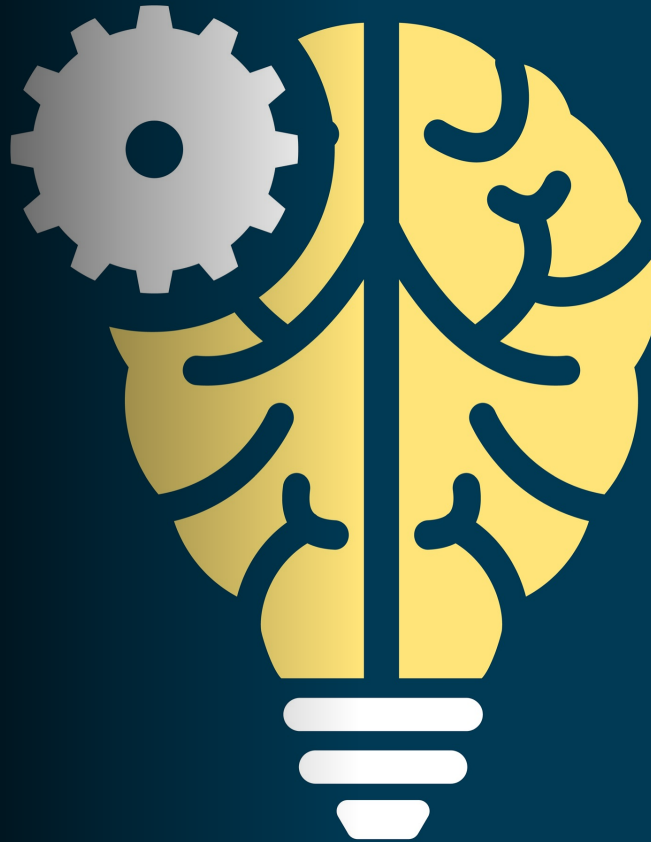
Dimensions:

X, Y, Z

Table and Depth

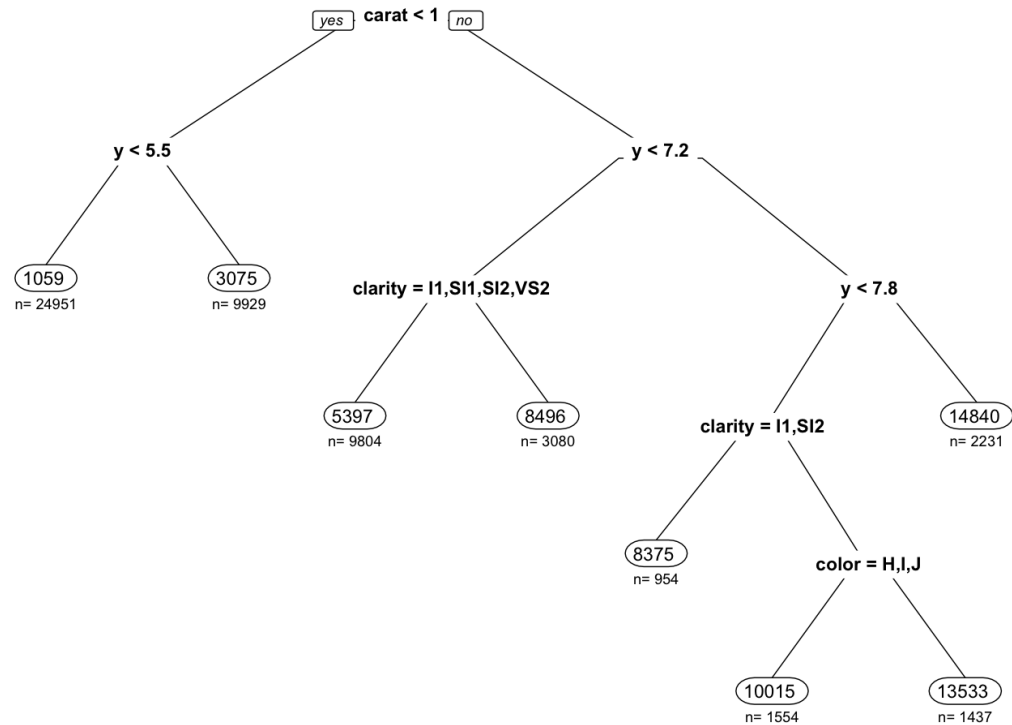
Why Use Machine Learning

- Non-linear pricing patterns
- Strong interactions
- Traditional models underperform
- Random Forest handles:
- Nonlinear effects
- High-order interactions
- Robustness on large datasets



How the Random Forest Works

- Builds hundreds of decision trees
- Each tree sees random subsets
- Final prediction = averaged trees
- Reduces overfitting
- Provides model-based variable importance

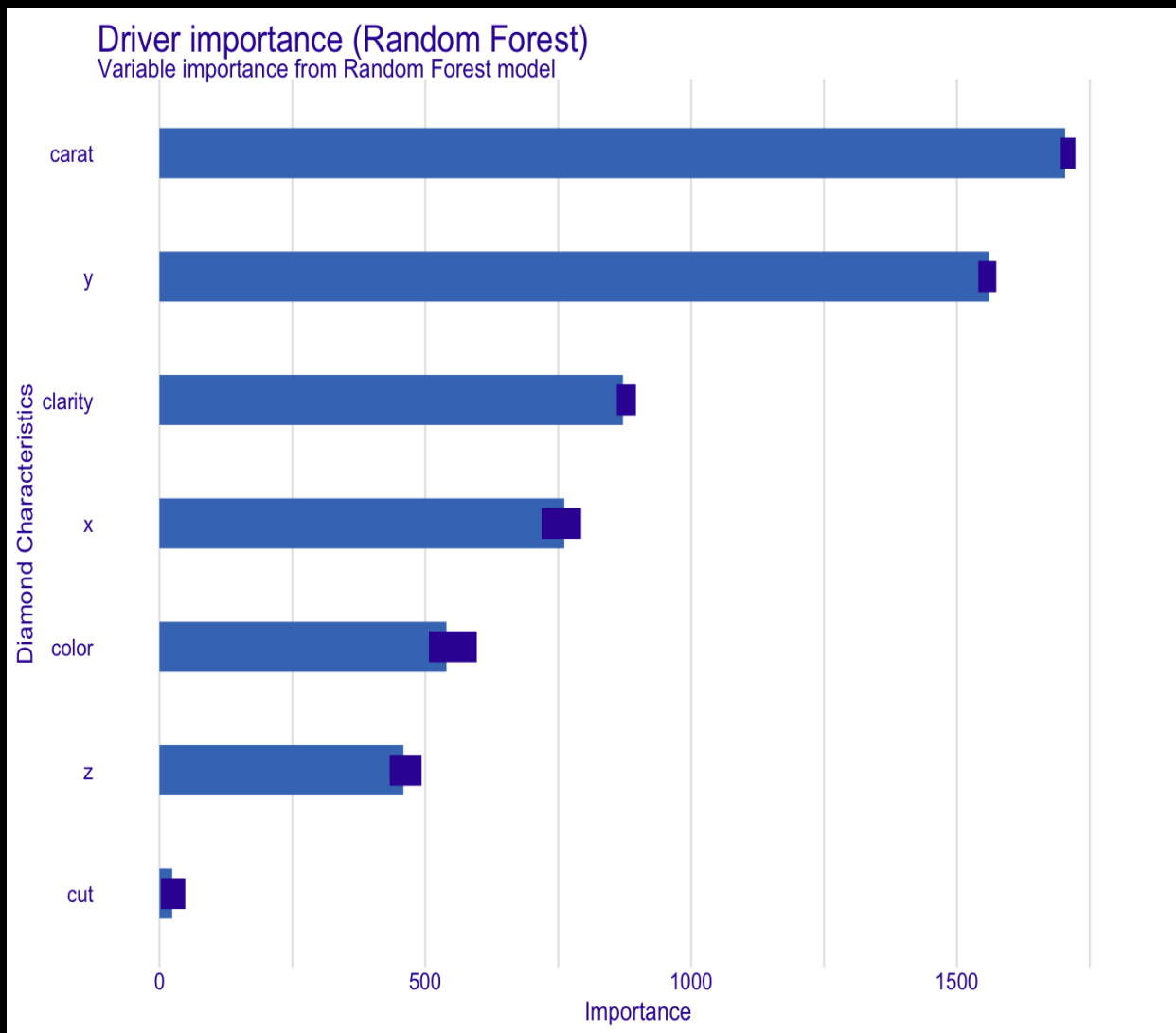


Model Performance

- **Naïve Model (predicting the average price)**
- **RMSE: \$3,994.09 - MAE: \$3,017.67**
- **Random Forest Model (our final model)**
- **RMSE: \$651.57 - MAE: \$328.78**
- **Improvement vs. Naïve Approach**
- RMSE reduced by $\approx 84\%$ - MAE reduced by $\approx 89\%$
- Model explains **97.4%** of variation in diamond prices



RANK	DRIVER VARIABLE	RELATIONSHIP TO PRICE	KEY TAKEAWAY
1	Carat	Strong positive - price rises exponentially as carat weight increases	Size is the primary determinant of value
2	X, Y, Z	The dimensions (mm) have a strong relationship with carat weight.	The larger the dimensions the higher the price
3	Clarity	Higher clarity grades (IF → VS1) have premium prices	Buyers pay for fewer flaws
4	Color	Price drops as color grade moves from D (best) to J (worst)	Color matters moderately but less than clarity
5	Cut Quality	Small differences between Ideal and Premium cuts	Cut affects appearance more than price



Variable
Importance

Carat vs. Price

- Strong nonlinear increase in price
- Fast growth from 0.5-1.5 carats
- Insight: Mid-carat stones (1.0–1.3 ct) offer best margin/value.



Clarity vs. Price

- Lower clarity grades (I1, SI2) show the highest predicted prices because they often appear in larger stones.

- Higher clarity grades (VS, VVS, IF) show lower average prices in this dataset.

- Carat dominates, overshadowing clarity's effect on price.

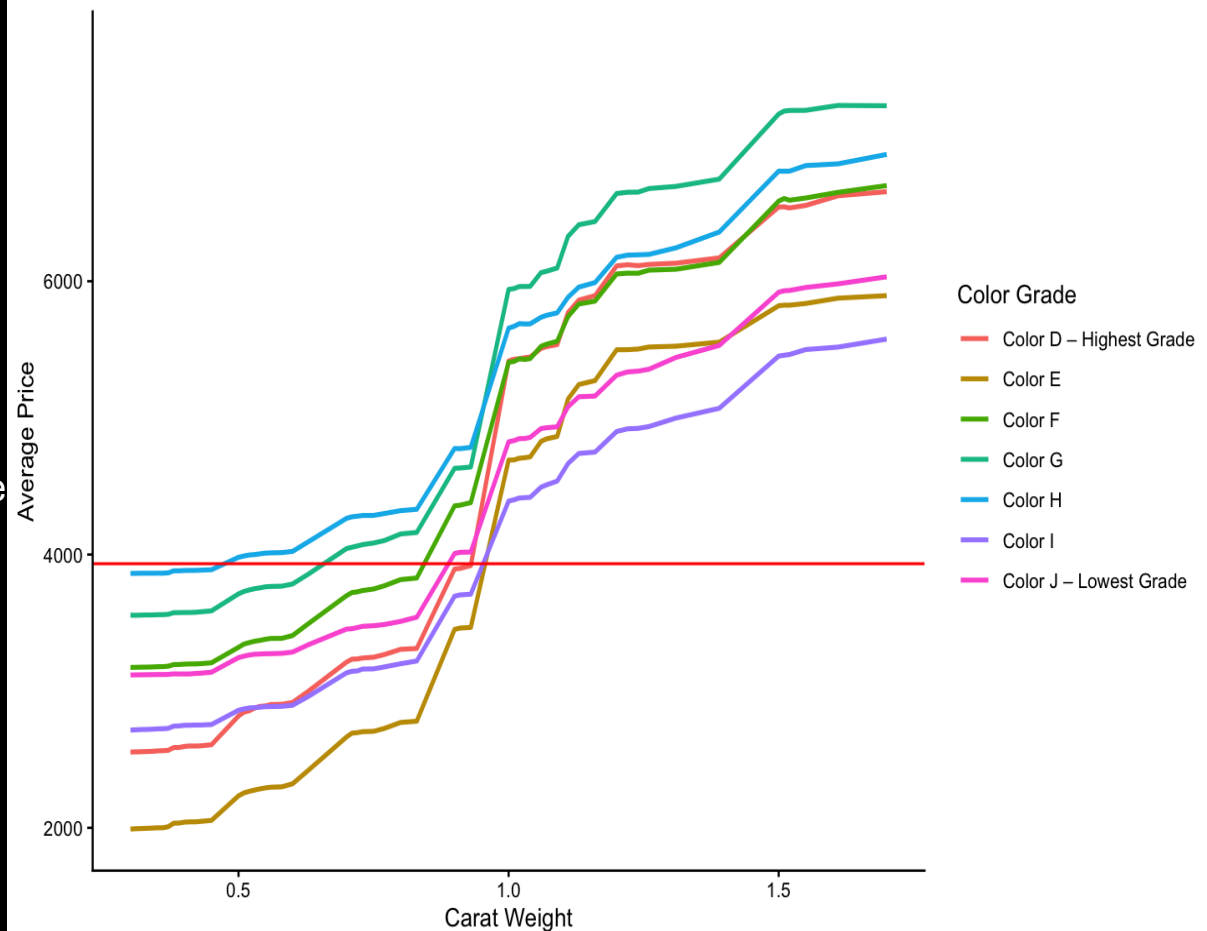


Carat vs. Color Interaction

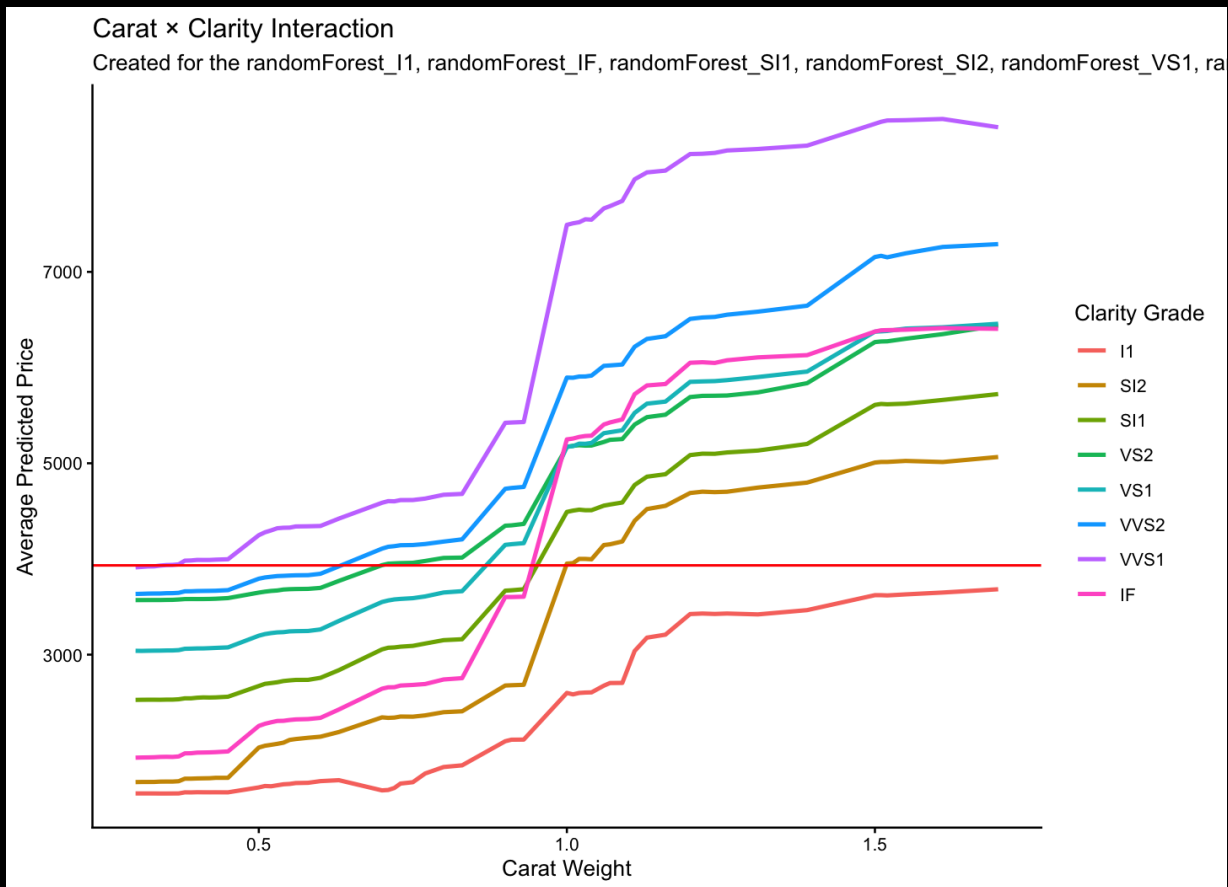
- Color differences matter most at smaller carat sizes (under ~0.9 ct).
- As carat increases, it dominates price even with optimal colors.
- For BrightStone: emphasize color on smaller diamonds, and emphasize carat on larger diamonds.

Color Differences Matter Most at Smaller Carat Sizes

For larger stones, carat dominates price regardless of color

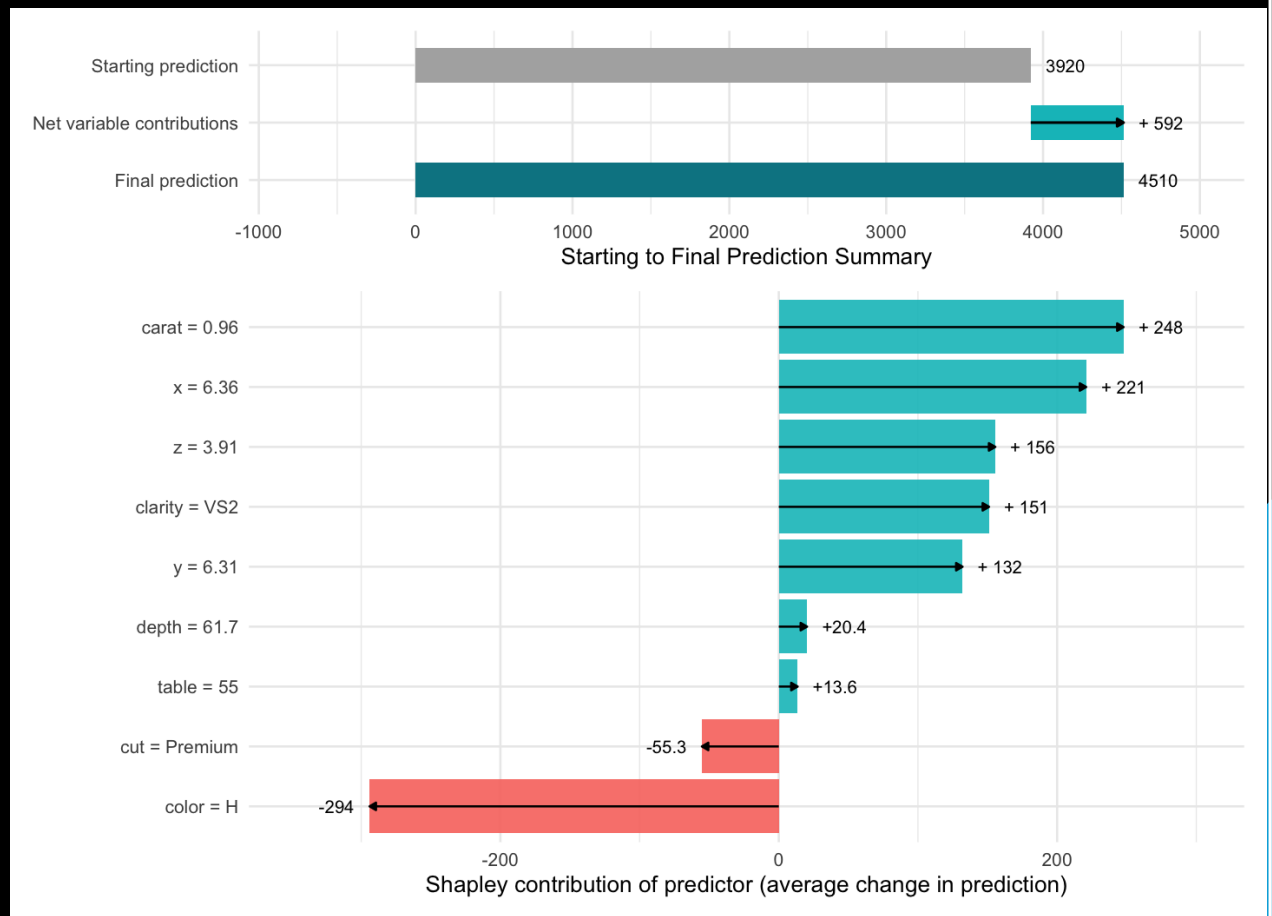


Carat vs. Clarity Interaction

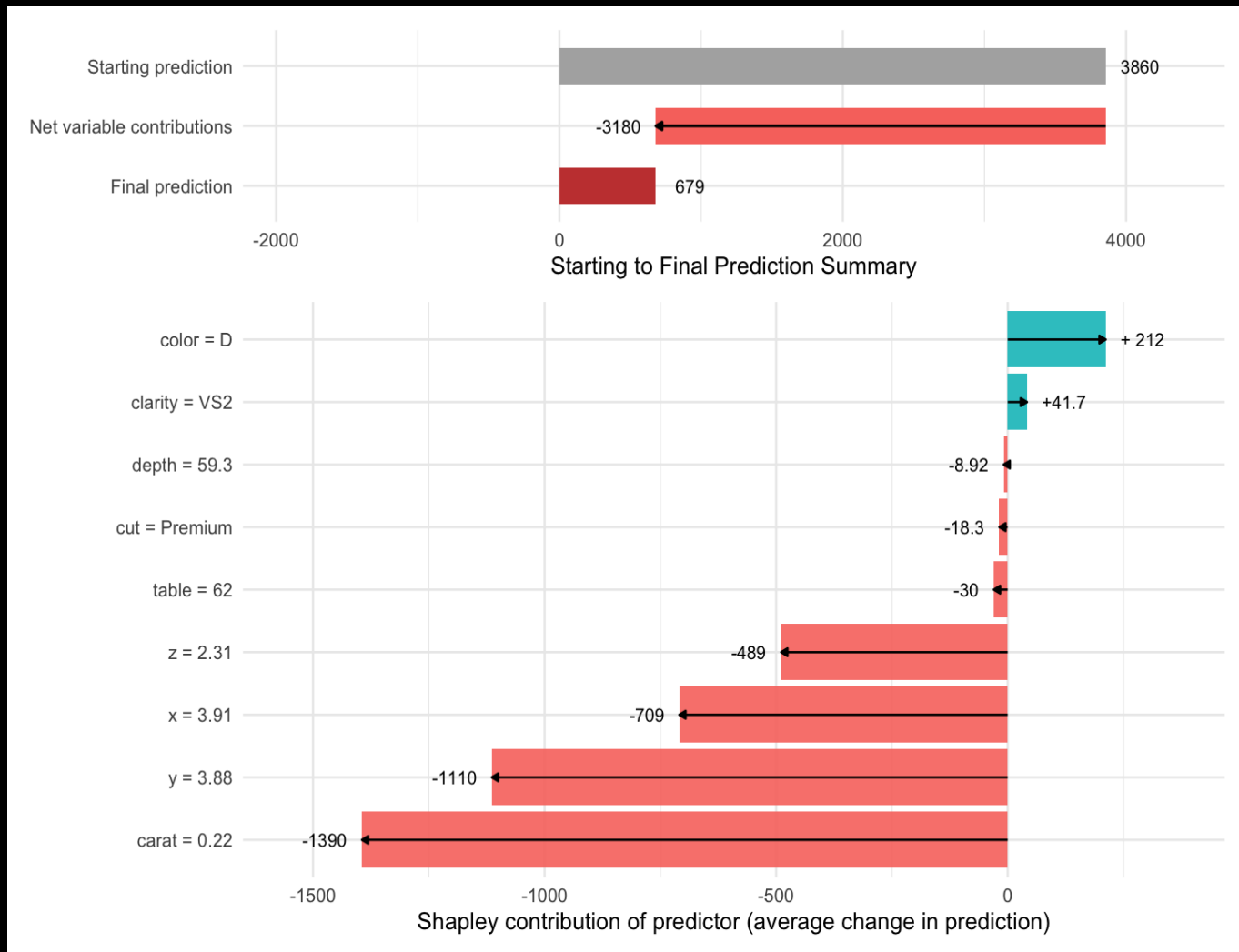


- Clarity differences matter most at smaller carat sizes (under ~0.9 ct).
- Some lower-clarity grades (like SI2/SI1) spike higher because they appear in larger stones in this dataset.
- For BrightStone: clarity influences price mainly on smaller stones, while carat drives price on larger stones.

High-Price Breakdown

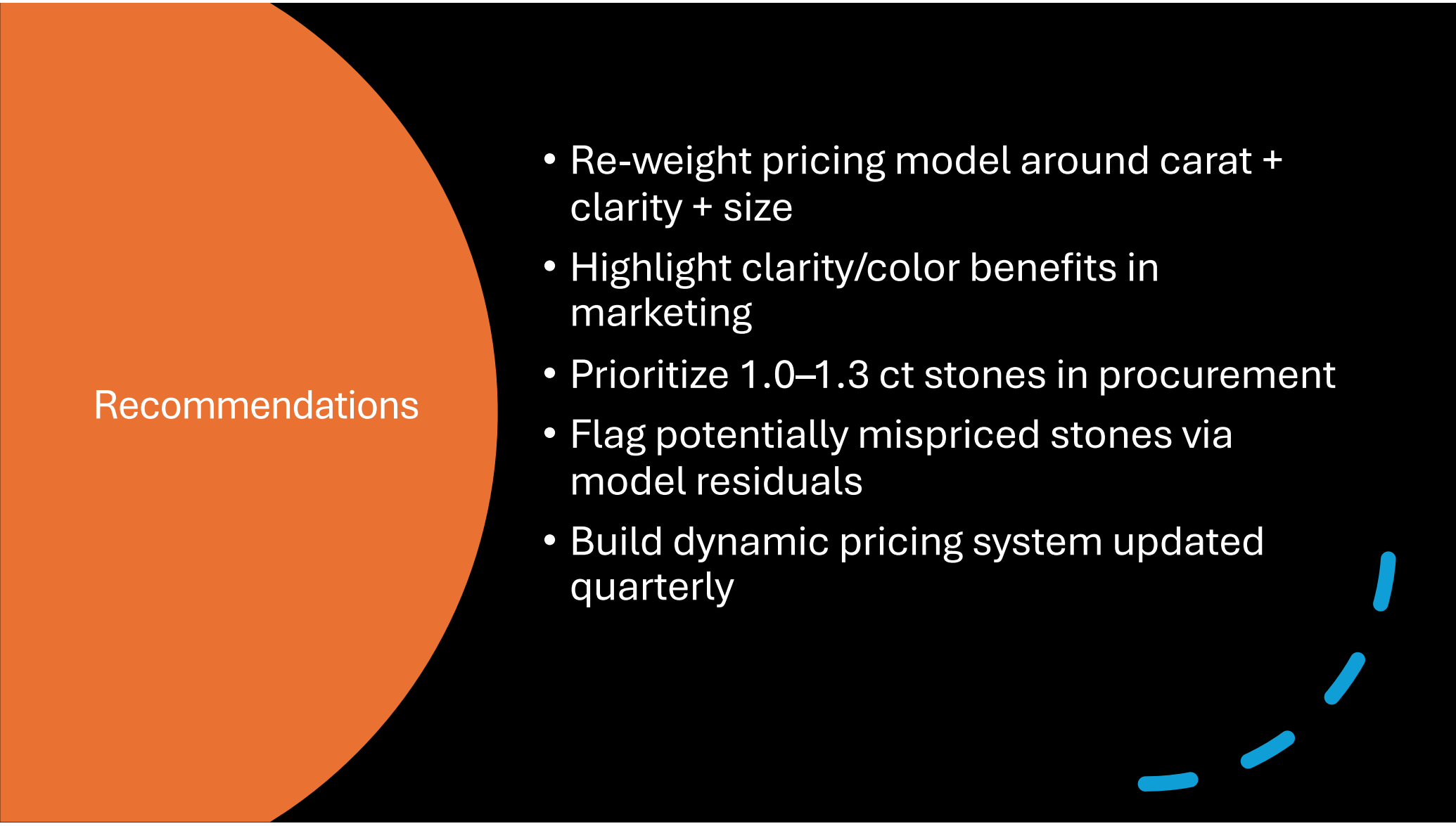


Low-Price Breakdown



Key Insights

- Carat is the #1 driver by a wide margin
- Clarity and color shape premium tiers
- Dimensions matter more than depth/table
- Interaction: size premium multiplies with quality levels



Recommendations

- Re-weight pricing model around carat + clarity + size
- Highlight clarity/color benefits in marketing
- Prioritize 1.0–1.3 ct stones in procurement
- Flag potentially mispriced stones via model residuals
- Build dynamic pricing system updated quarterly

Limitations



Key diamond attributes are missing (fluorescence, certification, and supplier differences), which limits price precision.



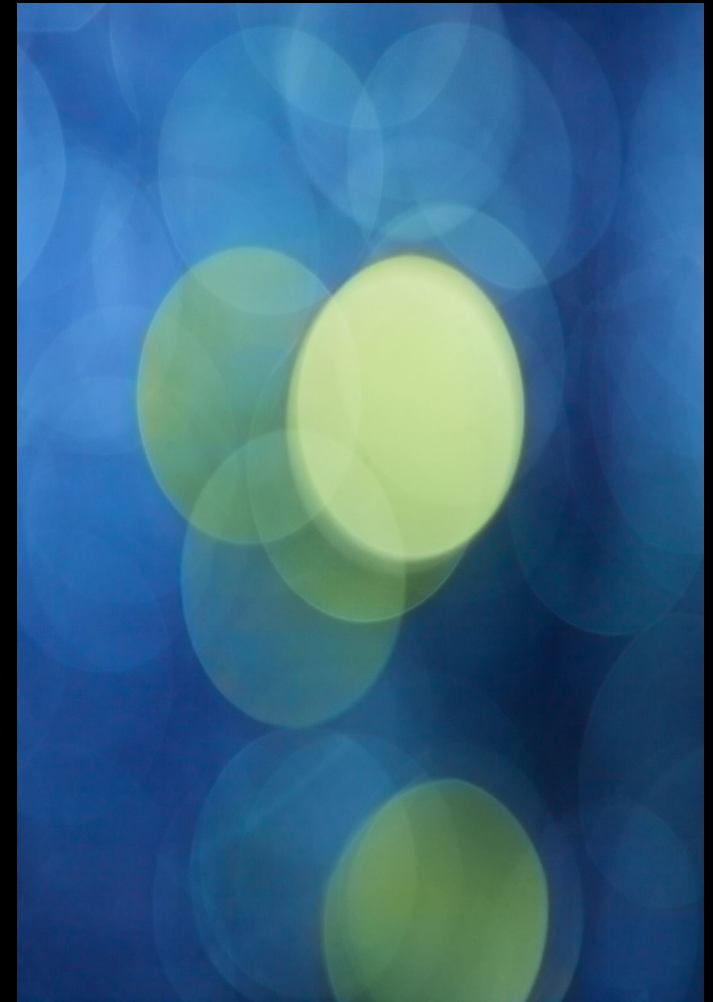
No cost or margin data, so insights focus on revenue potential rather than profitability.



Model reflects BrightStone's historical sales only and does not incorporate broader market trends or competitor pricing.



Random Forest model is a black box, requiring interpretability tools to translate predictions into business actions.



Conclusion



Gemstone Analytics provided BrightStone with:



Accurate predictive model



Clear driver understanding



Actionable pricing insights



Questions?